DOMINATING AN FPPF COVER BY A FINITE SURJECTIVE MORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2200-2207 of sources/stacks-stacks-more-morphisms/pinned-original-source.tex; proof span is lines 2209-2228, the complete author proof (18 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 45 of queue_stacks-stacks-more-morphisms.json (pb=858, stmt 2200-2207, proof 2209-2228).

Editorial note. stacks-stacks-more-morphisms wave smm01 (2026-10-14): fresh adjudication accept per lane-s3/smm_judgments.md; pool per Chief 10:5x open-pool ruling (net 59 with-proof rows, bloodline lane-s3 by anchor audit, v2 true-key namespace stacks:stacks-more-morphisms.tex, row lemma-deform-property-fp-over-ft avoided on cross-chapter key collision); 60 with-proof rows judged 45 accept / 14 reject / 1 avoid; patch-sd-envs contract spans scanned.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0CP0: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 2228–2228. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CNY: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 2228–2228. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 09YC: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 2228–2228. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let Y be a quasi-compact and quasi-separated algebraic space. Let $V \subset Y$ be a quasi-compact open. Let $f : \mathcal{X} \to V$ be surjective, flat, and locally of finite presentation. Then there exists a finite surjective morphism $g : Y' \to Y$ such that $V' = g^{-1}(V) \to Y$ factors Zariski locally through f.*

Date: stacks-project:stacks-more-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. We first prove this when Y is a scheme. We may choose a scheme U and a surjective smooth morphism $U \to \mathcal{X}$. Then $\{U \to V\}$ is an fppf covering of schemes. By More on Morphisms, Lemma 0CP0 there exists a finite surjective morphism $V' \to V$ such that $V' \to V$ factors Zariski locally through U. By More on Morphisms, Lemma 0CNY we can find a finite surjective morphism $Y' \to Y$ whose restriction to V is $V' \to V$ as desired.

If Y is an algebraic space, then we see the lemma is true by first doing a finite base change by a finite surjective morphism $Y' \to Y$ where Y' is a scheme. See Limits of Spaces, Proposition 09YC. □